Algebra I

expressions

A **term** is a number, a variable, or numbers and variables multiplied together: 5, x, and $-3x^2$ are all terms. An **expression** is terms added or subtracted.

Like terms have the same variable raised to the same power. Only like terms combine.

$3x$ and $-x$	like — both are plain x
$3x$ and $3x^2$	not like — different powers
5 and -2	like — both plain numbers

WORKED: SIMPLIFY $3x+5-x+2$
group the like terms: $(3x-x)+(5+2)$
combine each group: $2x+7$
($3x-x$ is $2x$, not 3. Removing one x from three x's leaves two.)

The **distributive rule** multiplies a bracket out: $a(b+c)=ab+ac$. Multiply the outside number by every term inside.

WORKED: $2(3x-4)$
$2\cdot 3x=6x$ and $2\cdot(-4)=-8$ $\Rightarrow$ $6x-8$

A minus sign outside a bracket flips every sign inside. $-(x-3)$ is $-x+3$, not $-x-3$.

To **evaluate** an expression, replace each letter with its number and follow the order of operations. Put negatives in brackets so the exponent applies correctly.

WORKED: $2x^2-3x$ AT $x=-2$
$2(-2)^2-3(-2)=2(4)+6=8+6=14$
$(-2)^2=4$, but $-2^2=-4$. The brackets decide whether the minus sign is squared.

solving linear equations

A **linear equation** has the variable to the first power only. Solving means getting the variable alone on one side.

THE PROCESS
1. **Clear brackets** with the distributive rule.
2. **Combine like terms** on each side.
3. **Move all variable terms** to one side, using the balance rule.
4. **Move all plain numbers** to the other side.
5. **Divide** by the number multiplying the variable.
6. **Check** by substituting into the original equation.

Undo operations in reverse order. The last thing done to the variable is the first thing you undo.

1. CLEAR THE BRACKETS
$3(x-2)+4=2x+9$ becomes $3x-6+4=2x+9$

↓

2. COMBINE LIKE TERMS
$-6+4=-2$, so $3x-2=2x+9$

↓

3. MOVE THE VARIABLES TOGETHER
subtract $2x$ from both sides: $x-2=9$

↓

4. MOVE THE NUMBERS
add 2 to both sides: $x=11$

↓

5. CHECK IN THE ORIGINAL
left: $3(11-2)+4=3(9)+4=31$ right: $2(11)+9=31$
Both sides give 31 ✓

turning words into algebra

Most people who find word problems hard are not bad at algebra. They are stuck at the translation, which is a separate skill worth practising on its own.

THE PROCESS
1. **Read it once** without writing anything.
2. **Name the unknown.** Write "let $x=$" and then the thing in words.
3. **Translate one phrase at a time** into symbols.
4. **Solve** the equation.
5. **Answer the question asked**, with units.

Step 2 is the one people skip. Writing "let $x=$ the number of tickets" fixes what every later symbol means.

the words	the symbol
is, was, gives, will be	$=$
sum, total, more than, increased by	$+$
difference, less than, decreased by	$-$
product, of, times, twice	$\times$
quotient, per, divided by, ratio	$\div$
a number, how many, what	the variable

"Less than" reverses the order. "Five less than a number" is $x-5$, not $5-x$. Read it as "take five away from the number". "More than" does not reverse, because addition may be written either way round.

WORKED: "FIVE LESS THAN TWICE A NUMBER IS 11"
name it: let $x=$ the number
translate in pieces: "twice a number" is $2x$ · "five less than" that is $2x-5$ · "is 11" is $=11$

$$2x-5=11$$

solve: $2x=16$, so $x=8$
check against the words, not the equation: twice 8 is 16, and five less than 16 is 11 ✓

inequalities

An **inequality** uses $<$, $>$, $\leq$ or $\geq$ instead of $=$. Solve it exactly like an equation, with one extra rule.

THE FLIP RULE
If you **multiply or divide both sides by a negative number**, reverse the inequality sign.

$$-2x>8 \quad\Longrightarrow\quad x<-4$$

Adding or subtracting a negative does not flip it. Only multiplying or dividing does.

WORKED: $-2x+3>11$
subtract 3 from both sides: $-2x>8$ *(no flip — this was subtraction)*
divide both sides by -2: $x<-4$ *(flip — this was division by a negative)*

Why the flip is necessary: $3<5$ is true. Multiply both sides by -1 and you get -3 and -5. Now $-3>-5$, so the sign must turn around to stay true.

Checking an inequality — test one value

An equation has one answer to check. An inequality has infinitely many, so you check a region instead.

THE TEST
1. Pick any easy number inside your answer, usually 0.
2. Put it into the original inequality.
3. It must be true. Then pick a number outside, which must be false.

WORKED: DID THE FLIP GO THE RIGHT WAY?
Answer claimed: $x<-4$. Test $x=-10$, which is inside:

$$-2(-10)+3=23 \quad \text{and} \quad 23>11 \ \checkmark$$

Now test $x=0$, which is outside: $-2(0)+3=3$, and $3>11$ is false ✓
Both results agree with the answer, so the flip went the right way. Had you forgotten the flip and written $x>-4$, then $x=0$ would have to work, and it does not.

This one habit catches the flip mistake every time. It costs about fifteen seconds.

Compound inequalities

A **compound inequality** traps the variable between two bounds: $-3<2x+1\le 7$. Whatever you do, do it to all three parts.

WORKED: $-3<2x+1\le 7$
subtract 1 from all three parts: $-4<2x\le 6$
divide all three by 2, a positive, so no flip: $-2<x\le 3$
check with $x=0$: $2(0)+1=1$, and $-3<1\le 7$ ✓

Note which end is included. The answer is $-2<x\le 3$, so 3 counts but -2 does not. The original signs carry through unchanged.

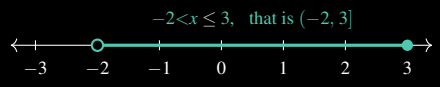
Interval notation

Interval notation writes an answer as the two endpoints. The bracket style says whether each endpoint is included.

symbol	means	example
[or]	included	$[2,5]$ is $2\le x\le 5$
(or)	excluded	$(2,5)$ is $2<x<5$
mixed	one of each	$(-2,3]$ is $-2<x\le 3$
∞	no end	always uses), never]

Infinity never gets a square bracket. You cannot include something you can never reach, so write $(3,\infty)$.

On a number line, a filled circle marks an included endpoint and an open circle an excluded one.



absolute value

Recall that $|x|$ is the distance from zero, so it is never negative. Distance forgets direction, and that is what makes these equations have two answers.

THE TWO-CASE RULE
$|A|=b$ with $b>0$ splits into two equations:

$$A=b \quad \text{or} \quad A=-b$$

Isolate the absolute value first. Split only when the bars stand alone on one side.

WORKED: $|x-3|=5$
split into two cases:
$x-3=5$ gives $x=8$. $x-3=-5$ gives $x=-2$.

$$x=8 \quad \text{or} \quad x=-2$$

check: $|8-3|=|5|=5$ ✓ and $|-2-3|=|-5|=5$ ✓

$|A|=$ a negative number has no solution. $|x|=-2$ is impossible, because a distance is never negative. Say so and stop.

Absolute value inequalities — less is between, greater is beyond

A $|A|<b \Rightarrow$ one compound inequality: $-b<A<b$. The answer is a single stretch around the centre.

B $|A|>b \Rightarrow$ two separate inequalities: $A>b$ **or** $A<-b$. The answer is two pieces heading outwards.

Read the meaning and you never have to memorize this. "Distance less than 5" means near the centre, so one stretch. "Distance more than 5" means far away on either side, so two pieces.

WORKED (A): $|x-3|<5$
distance from 3 is less than 5, so x sits within 5 of 3:

$$-5<x-3<5 \implies -2<x<8$$

in interval notation: $(-2,8)$
check $x=0$: $|0-3|=3$, and $3<5$ ✓

↓

WORKED (B): $|x-3|>5$
distance from 3 is more than 5, so x sits far out on either side:

$$x-3>5 \quad \text{or} \quad x-3<-5 \implies x>8 \ \text{or} \ x<-2$$

in interval notation: $(-\infty,-2)\cup(8,\infty)$
check $x=9$: $|9-3|=6$, and $6>5$ ✓ check $x=0$: $3>5$ is false, and 0 is correctly outside the answer ✓

Case B is never written as one chain. Writing $5<x-3<-5$ claims a number is both large and small at once, which nothing satisfies.

lines and slope

Slope measures steepness: how far the line rises for each step to the right.

$$m=\frac{y_2-y_1}{x_2-x_1}=\frac{\text{rise}}{\text{run}}$$

WORKED: SLOPE THROUGH $(2,3)$ **AND** $(5,7)$

$$m=\frac{7-3}{5-2}=\frac{4}{3}$$

Keep the points in the same order top and bottom. Starting with 7 on top means starting with 5 on the bottom.

$m>0$	rises left to right
$m<0$	falls left to right
$m=0$	horizontal — equation is $y=$ a number
undefined	vertical — equation is $x=$ a number

Three forms of a line

form	looks like	use it when
slope-intercept	$y=mx+b$	graphing
point-slope	$y-y_1=m(x-x_1)$	you have a point and m
standard	$Ax+By=C$	finding intercepts

In $y=mx+b$ the number b is the **y-intercept**, where the line crosses the y-axis.

WORKED: LINE THROUGH $(2,3)$ WITH $m=\frac{4}{3}$

start from point-slope: $y-3=\frac{4}{3}(x-2)$

distribute: $y-3=\frac{4}{3}x-\frac{8}{3}$

add 3, which is $\frac{9}{3}$: $y=\frac{4}{3}x+\frac{1}{3}$

parallel lines same slope, different intercept

perpendicular lines slopes are negative reciprocals: flip and change the sign

A line perpendicular to slope $\frac{4}{3}$ has slope $-\frac{3}{4}$. *(Multiply them: $\frac{4}{3}\cdot-\frac{3}{4}=-1$, which is the test.)*

systems of equations

A **system** is two equations that must both be true at once. The solution is the point where the two lines cross.

Which method? — look at the equations

A **substitution** — one equation already has a variable alone ($y=\dots$), or is easy to make so

B **elimination** — matching terms line up, so adding or subtracting cancels one variable

WORKED (A): $y=2x-1$ AND $3x+y=9$

y is already alone, so replace y in the second equation:

$3x+(2x-1)=9 \;\Rightarrow\; 5x-1=9 \;\Rightarrow\; 5x=10 \;\Rightarrow\; x=2$

put $x=2$ back into $y=2x-1$: $y=4-1=3$

solution $(2,\,3)$

↓

WORKED (B): $2x+3y=12$ AND $2x-y=4$

both have $2x$, so subtract the second from the first and the x terms cancel:

$$(2x-2x)+(3y-(-y))=12-4 \;\Rightarrow\; 4y=8 \;\Rightarrow\; y=2$$

put $y=2$ into $2x-y=4$: $2x-2=4 \;\Rightarrow\; x=3$

solution $(3,\,2)$

CHECK BOTH EQUATIONS, NOT ONE

$2(3)+3(2)=6+6=12$ ✓ $\quad 2(3)-2=4$ ✓

A point can satisfy one equation and fail the other. Checking only one proves nothing.

one solution the lines cross once

no solution parallel lines. You reach a false statement such as $0=5$.

infinitely many the same line twice. You reach $0=0$.

Systems of three equations

Three equations, three unknowns. The method is elimination, applied twice.

THE PROCESS

1. **Pick** one variable to remove.
2. **Eliminate** it from one pair of equations.
3. **Eliminate** the same variable from a different pair.
4. **Solve** the two-variable system you now have.
5. **Back-substitute** to find the third variable.

WORKED: $x+y+z=6$, $2x-y+z=3$, $x+2y-z=2$

eliminate z from the first pair by subtracting: $(x+y+z)-(2x-y+z)=6-3$

$$-x+2y=3$$

eliminate z from the first and third by adding: $(x+y+z)+(x+2y-z)=6+2$

$$2x+3y=8$$

solve those two: from the first, $x=2y-3$. Substitute: $2(2y-3)+3y=8$, so $7y=14$ and $y=2$.

then $x=2(2)-3=1$, and from the original first equation $z=6-1-2=3$

solution $(1,\,2,\,3)$

Check all three equations. $1+2+3=6$ ✓ $\quad 2-2+3=3$ ✓ $\quad 1+4-3=2$ ✓

Step 3 must use a different pair. Reusing the same two equations gives you back the same information, and the system will not resolve.

exponent rules

Every rule below comes from writing the powers out and counting.

multiply $x^a\cdot x^b=x^{a+b}$ $\quad x^3x^5=x^8$

divide $\frac{x^a}{x^b}=x^{a-b}$ $\quad \frac{x^8}{x^3}=x^5$

power of a power $(x^a)^b=x^{ab}$ $\quad (x^3)^5=x^{15}$

power of a product $(xy)^a=x^ay^a$

zero power $x^0=1$ for any $x\neq 0$

negative power $x^{-a}=\frac{1}{x^a}$

Why $x^3x^5=x^8$: write them out. Three x's times five x's is eight x's in a row. Why $x^0=1$: $\frac{x^3}{x^3}$ is 1, and the divide rule makes it x^0.

Add exponents when multiplying, multiply them when raising a power to a power. Swapping these two is the most common exponent error.

A negative exponent does not make the number negative. $2^{-3}=\frac{1}{8}$, which is positive. It means "one over", not "less than zero".

polynomials

A **polynomial** is a sum of terms whose variables have whole-number powers, such as $2x^2-5x-12$. The **degree** is the highest power.

To multiply two brackets, multiply every term in the first by every term in the second.

WORKED: $(2x+3)(x-4)$

four products, one per pair:

$$2x\cdot x=2x^2 \qquad 2x\cdot(-4)=-8x \qquad 3\cdot x=3x \qquad 3\cdot(-4)=-12$$

add them, then combine the like terms $-8x$ and $3x$:

$$2x^2-8x+3x-12=2x^2-5x-12$$

$(a+b)^2$ is not a^2+b^2. Multiply it out: $(a+b)^2=a^2+2ab+b^2$. The middle term is real and is the one people lose.

factoring

Factoring reverses multiplication. It rewrites a sum as a product, which is what lets you solve equations later.

ALWAYS DO THIS FIRST

Take out the greatest common factor. If every term shares something, pull it out before anything else.

$$6x^3+9x^2=3x^2(2x+3)$$

Skipping this makes every later step harder, and sometimes impossible.

Which method? — count the terms

terms	shape	method
2	a^2-b^2	difference of squares
2	$a^3\pm b^3$	sum or difference of cubes
3	x^2+bx+c	find two numbers
3	ax^2+bx+c, $a\neq 1$	the AC method
4	anything	grouping

difference of squares $a^2-b^2=(a+b)(a-b)$

perfect square $a^2\pm 2ab+b^2=(a\pm b)^2$

sum of cubes $a^3+b^3=(a+b)(a^2-ab+b^2)$

difference of cubes $a^3-b^3=(a-b)(a^2+ab+b^2)$

A sum of squares a^2+b^2 does not factor. Only the difference does.

WORKED (2 TERMS): $4x^2-25$

both parts are perfect squares: $4x^2=(2x)^2$ and $25=5^2$

$$4x^2-25=(2x+5)(2x-5)$$

↓

WORKED (3 TERMS, $a=1$): x^2+5x+6

find two numbers that multiply to 6 and add to 5

pairs for 6: 1 and 6 give 7. 2 and 3 give 5 ✓

$$x^2+5x+6=(x+2)(x+3)$$

↓

WORKED (3 TERMS, $a\neq 1$): $6x^2+11x+3$ **— THE AC METHOD**

1. multiply a and c: $6\cdot 3=18$

2. find two numbers that multiply to 18 and add to 11: 9 and 2

3. split the middle term using them: $6x^2+9x+2x+3$

4. group in pairs and factor each: $3x(2x+3)+1(2x+3)$

5. the bracket now repeats, so pull it out:

$$(3x+1)(2x+3)$$

Check any factoring by multiplying back out. It takes ten seconds and catches every sign error.

quadratic equations

A **quadratic equation** has an x^2 term and no higher power. Written in **standard form** it is

$$ax^2+bx+c=0$$

Get everything on one side first. Every method below assumes the equation equals zero.

Which method? — look at the equation

A **factoring** — the quadratic factors easily. Fastest when it works.

B **square root property** — there is no plain x term, or one side is already a perfect square

C **completing the square** — asked for it, or you need **vertex form**

D **quadratic formula** — always works. Use it when nothing factors.

Try A first, fall back to D. B and C are for the shapes that ask for them.

THE ZERO PRODUCT RULE — WHY FACTORING SOLVES

If two numbers multiply to 0, at least one of them is 0. Nothing else can happen.

So once $(x-2)(x-3)=0$, either $x-2=0$ or $x-3=0$.

This works for zero only. If $(x-2)(x-3)=6$, you may not say $x-2=6$. Move everything to one side first.

WORKED (A) FACTORING: $x^2-5x+6=0$

two numbers multiplying to 6, adding to -5: -2 and -3

$(x-2)(x-3)=0$, so $x-2=0$ or $x-3=0$

$$x=2 \quad \text{or} \quad x=3$$

↓

WORKED (B) SQUARE ROOT PROPERTY: $(x-3)^2=16$

take the square root of both sides, and keep both signs:

$$x-3=\pm 4$$

$x-3=4$ gives $x=7$. $x-3=-4$ gives $x=-1$.

$$x=7 \quad \text{or} \quad x=-1$$

Forgetting the negative root loses half the answer. A square erases the sign, so undoing it must restore both.

↓

WORKED (C) COMPLETING THE SQUARE: $x^2+6x-7=0$

1. move the plain number across: $x^2+6x=7$

2. halve the x coefficient and square it: $\left(\frac{6}{2}\right)^2=3^2=9$

3. add it to both sides: $x^2+6x+9=16$

4. the left side is now a perfect square: $(x+3)^2=16$

5. square root both sides: $x+3=\pm 4$, so $x=1$ or $x=-7$

↓

WORKED (D) THE FORMULA: $2x^2+3x-5=0$

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a} \qquad a=2,\ b=3,\ c=-5$$

1. the part under the root: $b^2-4ac=3^2-4(2)(-5)=9+40=49$

2. substitute: $x=\frac{-3\pm\sqrt{49}}{2(2)}=\frac{-3\pm 7}{4}$

3. split the $\pm$: $x=\frac{4}{4}=1$ or $x=\frac{-10}{4}=-\frac{5}{2}$

$-4ac$ with a negative c becomes addition. Two negatives made $+40$ here.

The discriminant — how many answers before you solve

The part under the root, b^2-4ac, is the **discriminant**. It tells you the number of real answers without finishing the problem.

$b^2-4ac>0$	two real answers
$b^2-4ac=0$	one real answer, a repeated root
$b^2-4ac<0$	no real answers — the graph misses the x-axis

Algebra II

rational expressions

A **rational expression** is one polynomial divided by another, such as $\frac{x^2-9}{x^2+x-6}$. Treat it exactly like a fraction, with one addition: watch the denominator.

The denominator may never be zero. Any value that would make it zero is **excluded**, even if it cancels later. State those values.

THE PROCESS — SIMPLIFY
1. **Factor** the top and the bottom completely.
2. **Note the excluded values** — whatever makes the original bottom zero.
3. **Cancel** factors the top and bottom share.

WORKED: SIMPLIFY $\frac{x^2-9}{x^2+x-6}$

factor both: $\frac{(x+3)(x-3)}{(x+3)(x-2)}$

excluded: the original bottom is zero at $x=-3$ and $x=2$

cancel the shared $(x+3)$:

$$\frac{x-3}{x-2}, \qquad x \neq -3,\ x \neq 2$$

The restriction $x \neq -3$ survives the cancelling. The original expression never had a value there, so neither does the simplified one.

Adding needs a common denominator, exactly as with numbers.

WORKED: $\frac{1}{x}+\frac{1}{x+1}$

the common denominator is $x(x+1)$

$$\frac{1}{x}\cdot\frac{x+1}{x+1}+\frac{1}{x+1}\cdot\frac{x}{x}=\frac{x+1}{x(x+1)}+\frac{x}{x(x+1)}$$

add the tops: $\frac{2x+1}{x(x+1)}$

Solving — and the extraneous root trap

Multiply both sides by the common denominator, solve, then check every answer against the original.

WORKED: $\frac{x}{x-3}=\frac{3}{x-3}+2$

multiply everything by $(x-3)$: $x=3+2(x-3)$

solve: $x=3+2x-6 \ \Rightarrow\ -x=-3 \ \Rightarrow\ x=3$

Now check. Put $x=3$ into the original: the denominator $x-3$ becomes 0.

No solution.

The algebra was correct. The answer it produced is not allowed.

Why this happens: multiplying by $(x-3)$ is only legal when $x \neq 3$. The step quietly assumed what it then produced.

Complex fractions

A **complex fraction** has a fraction inside a fraction. Rewrite the main bar as a division and use the divide rule.

$$\frac{\frac{a}{b}}{\frac{c}{d}}=\frac{a}{b}\div\frac{c}{d}=\frac{a}{b}\cdot\frac{d}{c}=\frac{ad}{bc}$$

WORKED: $\frac{\frac{2}{3}}{\frac{4}{5}}$

$$\frac{2}{3}\cdot\frac{5}{4}=\frac{10}{12}=\frac{5}{6}$$

The long bar is the main division. When the layout is unclear, the longer bar always separates the top of the whole fraction from the bottom.

radicals and rational exponents

SIMPLIFYING A SQUARE ROOT
1. **Find** the largest perfect square that divides the number.
2. **Split** the root into two roots.
3. **Take** the perfect square out.

$$\sqrt{50}=\sqrt{25\cdot 2}=\sqrt{25}\cdot\sqrt{2}=5\sqrt{2}$$

Roots do not split across addition. $\sqrt{9+16}=\sqrt{25}=5$, but $\sqrt{9}+\sqrt{16}=3+4=7$. They split across multiplication only.

A **rational exponent** is another way to write a root:

$$x^{1/n}=\sqrt[n]{x} \qquad\qquad x^{m/n}=\sqrt[n]{x^m}$$

The bottom of the fraction is the root, the top is the power. So $8^{2/3}$ is the cube root of 8, squared: $2^2=4$.

Rationalizing clears a root out of a denominator. Multiply the top and bottom by that root.

$$\frac{3}{\sqrt{2}}=\frac{3}{\sqrt{2}}\cdot\frac{\sqrt{2}}{\sqrt{2}}=\frac{3\sqrt{2}}{2}$$

Solving radical equations — always check

1. ISOLATE THE ROOT, THEN SQUARE BOTH SIDES

$\sqrt{x+5}=x-1$ becomes $x+5=(x-1)^2=x^2-2x+1$

↓

2. SOLVE THE RESULT

$x^2-3x-4=0 \ \Rightarrow\ (x-4)(x+1)=0 \ \Rightarrow\ x=4$ or $x=-1$

↓

3. CHECK BOTH IN THE ORIGINAL — THIS STEP IS MANDATORY

$x=4$: $\sqrt{9}=3$ and $4-1=3$ ✓

$x=-1$: $\sqrt{4}=2$ but $-1-1=-2$. $2 \neq -2$, so reject it

$x=4$ only

Why squaring invents answers: squaring turns $2=-2$ into the true statement $4=4$. It erases sign differences, so it can make a false equation look true. The check undoes that damage.

complex numbers

Some equations have no real answer, because no real number squares to a negative. Mathematicians defined one:

$$i=\sqrt{-1} \qquad \text{so} \qquad i^2=-1$$

A **complex number** has the form $a+bi$, with a the **real part** and b the **imaginary part**.

$\sqrt{-9}=\sqrt{9}\cdot\sqrt{-1}=3i$

add — combine real with real, imaginary with imaginary

multiply — multiply out as brackets, then replace i^2 with -1

WORKED: $(3+2i)+(1-5i)$

reals: $3+1=4$. imaginaries: $2i-5i=-3i$.

$4-3i$

↓

WORKED: $(3+2i)(1-5i)$
multiply all four pairs:

$$3\cdot1=3 \quad 3\cdot(-5i)=-15i \quad 2i\cdot1=2i \quad 2i\cdot(-5i)=-10i^2$$

replace i^2 with -1: $-10i^2=-10(-1)=+10$
collect: $(3+10)+(-15i+2i)=13-13i$
The i^2 term becomes a real number. That is why the real part changed from 3 to 13.

exponential and logarithmic functions

An **exponential function** puts the variable in the exponent: $f(x)=2^x$. It grows by multiplying by the same factor each step, not by adding.

A **logarithm** answers the question an exponential asks backwards.

$$\log_b(y)=x \qquad \text{means exactly} \qquad b^x=y$$

Read $\log_2(8)=3$ as "2 raised to what power gives 8?" The answer is 3.

A logarithm is an exponent. That single sentence removes most of the confusion in this topic.

product	$\log(xy)=\log x+\log y$
quotient	$\log\left(\frac{x}{y}\right)=\log x-\log y$
power	$\log(x^n)=n\log x$
change of base	$\log_b x=\frac{\log x}{\log b}$
inverses	$\log_b(b^x)=x$ and $b^{\log_b x}=x$

$\log(x+y)$ does not simplify. The product rule turns multiplication into addition, not addition into anything.

Which method? — solving these two shapes

A **variable in the exponent** ⇒ take a logarithm of both sides
B **variable inside a log** ⇒ combine into one log, then rewrite as an exponential

WORKED (A), THE EASY CASE: $2^x=32$
write 32 as a power of the same base: $32=2^5$
same base means the exponents match: $x=5$
(Always look for a common base first. It avoids the calculator entirely.)

↓

WORKED (A), THE GENERAL CASE: $3^x=20$
20 is not a power of 3, so take a log of both sides:

$$\log(3^x)=\log 20 \;\Rightarrow\; x\log 3=\log 20 \;\Rightarrow\; x=\frac{\log 20}{\log 3}\approx 2.727$$

The power rule is what frees the x: it moves the exponent down in front.

↓

WORKED (B): $\log_2 x+\log_2(x-2)=3$
1. combine with the product rule: $\log_2\big(x(x-2)\big)=3$
2. rewrite as an exponential: $x(x-2)=2^3=8$
3. solve: $x^2-2x-8=0 \;\Rightarrow\; (x-4)(x+2)=0 \;\Rightarrow\; x=4$ or $x=-2$
4. check both. $x=-2$ gives $\log_2(-2)$, and a logarithm of a negative number does not exist.
$x=4$ only

Log equations almost always produce a rejected answer. Checking is part of the method, not an optional extra.

growth and decay

Exponential functions model anything that changes by a percentage rather than a fixed amount.

A **growth** ⇒ $A=A_0(1+r)^t$, with r the rate as a decimal
B **decay** ⇒ $A=A_0(1-r)^t$
C **compound interest** ⇒ $A=P\left(1+\frac{r}{n}\right)^{nt}$, with n compoundings a year
D **half-life** ⇒ $A=A_0\left(\frac{1}{2}\right)^{t/h}$, with h the half-life

Write the rate as a decimal, not a percent. Five percent is 0.05, so the base is 1.05 and not 1+5.

WORKED: \$1000 AT 5% A YEAR FOR 10 YEARS

$$A=1000(1+0.05)^{10}=1000(1.05)^{10}\approx \$1628.89$$

The gain is \$628.89, not \$500. Simple interest would add 5% of the original each year. Compounding adds 5% of the current balance, so each year's gain is larger than the last.

To solve for the time t, the unknown sits in the exponent, so take a logarithm of both sides. That is case A of the log-solving table above.

sequences and series

A **sequence** is an ordered list of numbers. A **series** is what you get by adding them.

A **arithmetic** — each term adds a fixed amount d, the **common difference**
B **geometric** — each term multiplies by a fixed amount r, the **common ratio**

Tell them apart by subtracting and dividing. If consecutive differences match, it is arithmetic. If consecutive ratios match, it is geometric.

	arithmetic	**geometric**
term	$a_n=a_1+(n-1)d$	$a_n=a_1r^{n-1}$
sum	$S_n=\frac{n}{2}(a_1+a_n)$	$S_n=\frac{a_1(1-r^n)}{1-r}$

The formulas use $n-1$, not n. The first term needs no steps, so reaching term n takes $n-1$ of them.

WORKED (A): 3, 7, 11, … **— FIND** a_{10} **AND** S_{10}
differences are 4 each time, so $d=4$ and $a_1=3$

$$a_{10}=3+(10-1)(4)=3+36=39$$

$$S_{10}=\frac{10}{2}(3+39)=5(42)=210$$

↓

WORKED (B): 2, 6, 18, … **— FIND** a_6 **AND** S_5
ratios are 3 each time, so $r=3$ and $a_1=2$

$$a_6=2\cdot3^{6-1}=2\cdot243=486$$

$$S_5=\frac{2\left(1-3^5\right)}{1-3}=\frac{2(1-243)}{-2}=\frac{2(-242)}{-2}=242$$

sigma notation and the binomial theorem

Sigma notation is shorthand for a sum. The symbol $\sum$ says "add up", and the labels say where to start and stop.

$$\sum_{i=1}^{4} 3i = 3(1)+3(2)+3(3)+3(4) = 3+6+9+12 = 30$$

index	the letter under $\sum$, here i — a counter, not an unknown
lower limit	where the counter starts
upper limit	where it stops, and it is included

The index is not part of the answer. It counts the terms and then disappears, exactly like the counter in a loop.

Pascal's triangle and the binomial theorem

Expanding $(a+b)^n$ by multiplying out is slow. The coefficients follow a pattern instead.

$$\begin{array}{ll} n=0 & 1 \\ n=1 & 1\quad 1 \\ n=2 & 1\quad 2\quad 1 \\ n=3 & 1\quad 3\quad 3\quad 1 \\ n=4 & 1\quad 4\quad 6\quad 4\quad 1 \end{array}$$

Each entry is the sum of the two above it. The edges are always 1.

THE PROCESS — EXPAND $(a+b)^n$

1. Take row n of the triangle as the coefficients.
2. Start at $a^n b^0$ and step down: the power of a falls, the power of b rises.
3. Check that every term's powers add to n.

WORKED: $(x+2)^3$

row 3 is 1, 3, 3, 1, and here $a=x$, $b=2$:

$$1x^3+3x^2(2)+3x(2)^2+1(2)^3=x^3+6x^2+12x+8$$

$$x^3+6x^2+12x+8$$

Raise the whole b term, including its number. The third term is $3x(4)$, not $3x(2)$.

direct and inverse variation

Two quantities often move together in one of two fixed patterns. Both hide a **constant of variation** k that you find from one known pair.

A **direct** — "y varies directly as x" $\Rightarrow y=kx$. As one grows, so does the other.

B **inverse** — "y varies inversely as x" $\Rightarrow y=\frac{k}{x}$. As one grows, the other shrinks.

THE PROCESS — THE SAME THREE STEPS EITHER WAY

1. Write the correct form, $y=kx$ or $y=\frac{k}{x}$.
2. Find k by substituting the pair you were given.
3. Rewrite with k filled in, then answer the question.

WORKED (A): y VARIES DIRECTLY AS x, AND $y=12$ WHEN $x=3$

$y=kx$, so $12=k(3)$, giving $k=4$
the rule is $y=4x$, so at $x=5$: $y=4(5)=20$

↓

WORKED (B): y VARIES INVERSELY AS x, AND $y=6$ WHEN $x=2$

$y=\frac{k}{x}$, so $6=\frac{k}{2}$, giving $k=12$
the rule is $y=\frac{12}{x}$, so at $x=4$: $y=\frac{12}{4}=3$

Check the direction against common sense. In A the answer grew with x, and in B it shrank. If yours moves the wrong way, you picked the wrong form.

polynomial division

THE REMAINDER THEOREM

To find the remainder when $P(x)$ is divided by $(x-c)$, just evaluate $P(c)$. No division needed.
Consequence: if $P(c)=0$, the remainder is zero, so $(x-c)$ is a factor. This is how you find roots of a cubic.

WORKED: $P(x)=x^3-2x^2+3x-4$ DIVIDED BY $(x-2)$

the divisor is $x-2$, so $c=2$:

$$P(2)=2^3-2(2)^2+3(2)-4=8-8+6-4=2$$

The remainder is 2. Because it is not zero, $(x-2)$ is not a factor.
(The full division gives x^2+3 with remainder 2, which agrees.)

Match the sign of c to the divisor. Dividing by $(x-2)$ means $c=+2$. Dividing by $(x+2)$ means $c=-2$.

finding polynomial roots

Factoring works when the factors are obvious. For a cubic or higher, you need a way to generate candidates first.

THE RATIONAL ROOT THEOREM

Every rational root of a polynomial has the form

$$\pm\frac{\text{a factor of the constant term}}{\text{a factor of the leading coefficient}}$$

That list is finite, so you can test it. It lists candidates, not answers — most of them fail.

THE PROCESS

1. List the candidates from the theorem.
2. Test them with the remainder theorem until one gives 0.
3. Divide the polynomial by that factor.
4. Factor what is left, which is now one degree lower.

WORKED: $2x^3-3x^2-3x+2$

1. constant 2 gives $\pm1, \pm2$. leading coefficient 2 gives $\pm1, \pm2$.
candidates: $\pm1,\ \pm2,\ \pm\frac{1}{2}$
2. test them: $P(1)=2-3-3+2=-2$ no · $P(2)=16-12-6+2=0$ ✓
3. so $(x-2)$ is a factor. Dividing leaves $2x^2+x-1$.
4. factor that: $2x^2+x-1=(2x-1)(x+1)$

$$2x^3-3x^2-3x+2=(x-2)(2x-1)(x+1)$$

roots: $x=2,\ \frac{1}{2},\ -1$ — all three were on the candidate list ✓

Legend

Colours: what we tell you
orange — a new word
teal — the answer
red — a common mistake
white — explanation
boxes stack top to bottom
↓ means "next step"
Colours: inside the maths
blue — a letter x, y
green — a number $3, 7$
yellow — a function f, g
pink — brackets ( )
grey — signs $+ - =$
Comparing
$=$ is equal to
$\neq$ is not equal to
$<$ is less than
$>$ is greater than
$\leq$ less than or equal
$\geq$ greater than or equal
$\approx$ is about
Operations
$\cdot$ times
$\pm$ plus or minus
$\sqrt{\ }$ square root
$\sqrt[n]{\ }$ nth root
$|x|$ absolute value
x^{-1} one over x
f^{-1} inverse function
Names
π about 3.14159
θ an angle
i $\sqrt{-1}$
m slope
a_n the nth term
$\Rightarrow$ therefore